Collective Semantic Memory: Merge, Trust, and Staleness for Peer Memory in Multi-Agent Navigation

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Abstract

Problem. Collective memory is often treated as monotone: more shared information should help. In cooperative navigation with scarce semantic targets, this intuition fails. Fast global sharing improves target materialization but causes agents to race for the same targets; slow sharing avoids contention but leaves agents with stale or incomplete memories. **Approach.** We instantiate four memory topologies over the same symbolic-semantic substrate: independent private memories, a shared event bus, a central consensus aggregator, and peer-to-peer broadcast with cadence K. We then introduce a minimal *collective semantic memory* (CSM): peer broadcast plus explicit *merge*, *trust*, and *staleness* rules. Q/R/M/C stage rates are used only as measurement instrumentation, not as the contribution of this paper. **Findings.** On a 35,640-run multi-agent sweep, the fixed-cadence peer family exhibits a bottleneck shift: fast broadcast saturates materialization but collapses completion, while slow broadcast has the opposite profile. Mean time-to-success has an interior optimum at $K{=}8$. On a 3,240-run scarcity benchmark, the minimal CSM strictly dominates five fixed-cadence peer baselines on success rate: CSM reaches 0.617 with 95% CI $[0.610, 0.624]$, above every peer-K upper CI. **Scope.** Experiments are grid-world substrates. The claim is not that this is a complete theory of collective memory; it is a reproducible architecture split showing that merge/trust/staleness are the right explicit control surfaces after cadence alone is exhausted.

1 Introduction

Many multi-agent systems share information either by writing to a single global memory, by sending learned messages, or by periodically exchanging local state. These mechanisms are usually evaluated end-to-end. If success improves, communication is judged useful; if it does not, the reason is opaque. This paper studies a narrower architectural question: when multiple agents maintain semantic memory about places, what should be shared, at what rate, and with what weight?

The answer is not "share everything as soon as possible." In a scarcity setting, where N agents compete for $T < N$ semantic targets, fresh global memory can make all agents select the same target. The memory is then accurate but unhelpful. Conversely, no communication prevents herding but starves agents of useful discoveries made by peers. The design problem is therefore a trade-off between semantic freshness and occupancy contention.

We make three contributions.

1. **Architecture split.** We implement and compare four memory topologies over the same symbolic substrate: independent, shared-bus, central-aggregator, and peer-to-peer broadcast.

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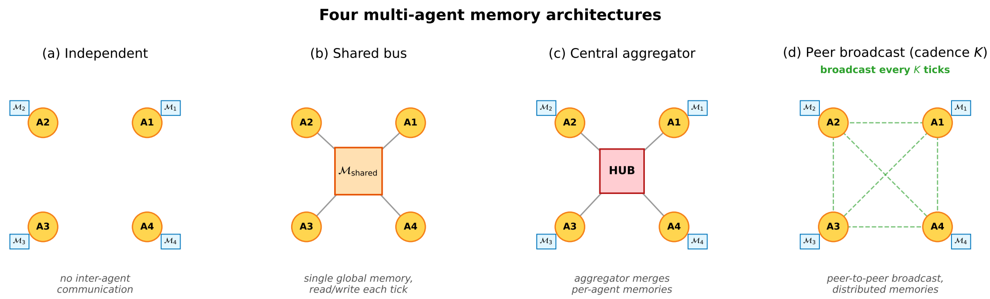

Figure 1: Four memory topologies. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads and writes each tick. **(c)** Central aggregator: private memories plus a hub that merges snapshots into a consensus report. **(d)** Peer-to-peer broadcast: private memories with snapshot exchange every K ticks and per-agent merged views.

2. **Cadence result.** We show that peer broadcast exposes a real coordination parameter K: fast broadcast improves $P(M^\star|R^\star)$ but hurts $P(C^\star|M^\star)$; the efficiency metric has an interior optimum.
3. **Minimal CSM.** We instantiate collective semantic memory with three explicit rules: merge, trust, and staleness. The minimal CSM reaches a new point on the M×C plane and strictly dominates fixed-cadence peer baselines on success rate in the scarcity benchmark.

Relation to the companion Q/R/M/C paper. The companion paper treats Q/R/M/C as a diagnostic framework. Here those stage rates are used as measuring instruments. The main object is different: the design of distributed symbolic memory itself.

2 Memory substrate and task

Each agent observes a grid environment and emits semantic place evidence such as `water_source`, `hazard_edge`, and `open`. A local memory stores place records, tag confidences, visit counts, freshness, and transition statistics. Planner queries retrieve candidate places by matching required, preferred, and penalty tags. A selected candidate is materialized into a concrete cell and executed by a local controller.

The multi-agent task uses a 12×10 grid. Each agent must reach a cell tagged `water_source`. Targets are scarce: $N\in\{3,5,8\}$ agents, $T\leq N$ targets, three layouts, three hazard densities, and 40 seeds for the main cadence sweep. Every run logs Q/R/M/C events per agent:

$$Q^\star \rightarrow R^\star \rightarrow M^\star \rightarrow C^\star,$$

where $M^\star$ means a semantic candidate was locked as an actionable target and $C^\star$ means the target was reached. These logs are not the architectural contribution; they let us see where each topology fails.

3 Four collective-memory topologies

The four topologies share the same local memory and planner. They differ only in where merge happens.

Independent. Each agent owns a private event store and concept graph. There is no communication path.

Shared bus. All observations are appended to a single global event bus. One shared concept graph is rebuilt from the union of all events.

Central aggregator. Each agent owns a private concept graph; a central consensus layer pulls snapshots and computes one trust-weighted report for everyone.

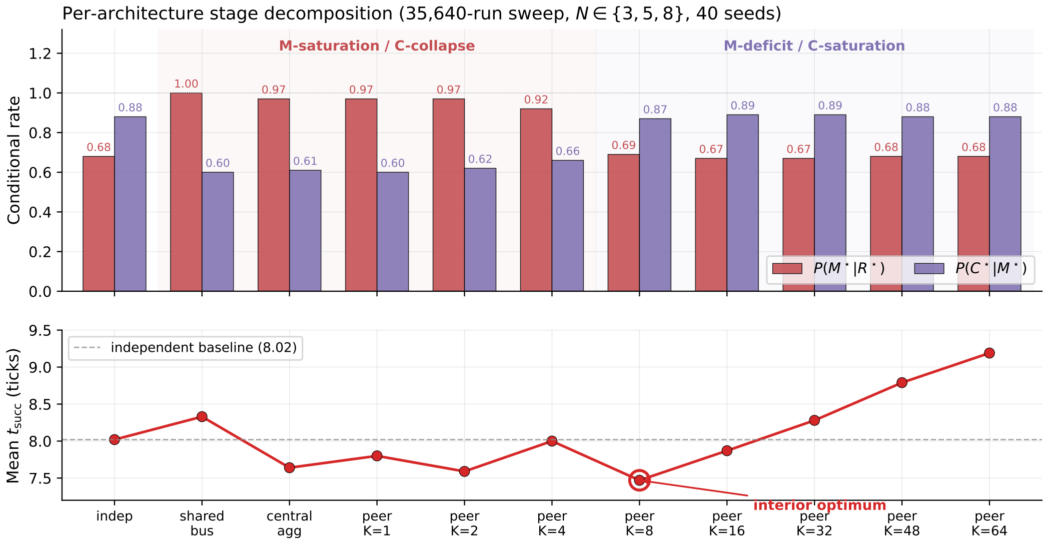

Figure 2: Conditional rates and mean time-to-success from the 35,640-run sweep. Fast sharing improves target materialization but harms completion through contention; slow sharing has the opposite profile. Mean time-to-success has an interior optimum at $K{=}8$.

Peer broadcast. Each agent owns a private concept graph and a private merged view. Every K ticks, agents broadcast snapshots; each agent merges its own memory with the latest snapshot from each peer.

The key distinction is not the clustering primitive. It is the level at which that primitive is instantiated:

$$\begin{array}{rcl} \text{independent} & : & N \text{ graphs, } 0 \text{ mergers} \\ \text{shared bus} & : & 1 \text{ graph, } 1 \text{ implicit global merger} \\ \text{central} & : & N \text{ graphs } + 1 \text{ central merger} \\ \text{peer} & : & N \text{ graphs } + N \text{ private mergers.} \end{array}$$

4 Cadence exposes the failure mode

The peer topology has one explicit parameter: broadcast cadence K. The main sweep crosses four architectures and eight peer cadences $K{\in}\{1, 2, 4, 8, 16, 32, 48, 64\}$ for 35,640 runs. Figure 2 summarizes the result.

Fast peer broadcast and centralized sharing live in the M-saturation/C-collapse regime: agents can lock targets reliably, but many locks point to targets that peers also pursue. Independent and slow peer broadcast live in the M-deficit/C-saturation regime: fewer agents lock onto good targets, but those that do are less likely to collide with peer occupancy.

Across all peer runs, Spearman of $P(M^\star|R^\star)$ vs K is -0.608 and of $P(C^\star|M^\star)$ vs K is $+0.420$ (both $p < 10^{-4}$). The stronger signature is visible at fixed cadence: the within-cadence Pearson correlation between the M and C links peaks at -0.61 for $K{=}4$ and decays toward noise as K grows. Cadence therefore does not simply tune "amount of communication." It moves the bottleneck between semantic commitment and shared-resource completion.

5 Minimal collective semantic memory

The cadence sweep shows that fixed-rate snapshot exchange exposes only one control surface. A collective semantic memory should expose at least three:

Merge. How peer evidence about the same semantic place is combined.

Trust. How evidence from different peers is weighted when it conflicts or becomes unreliable.

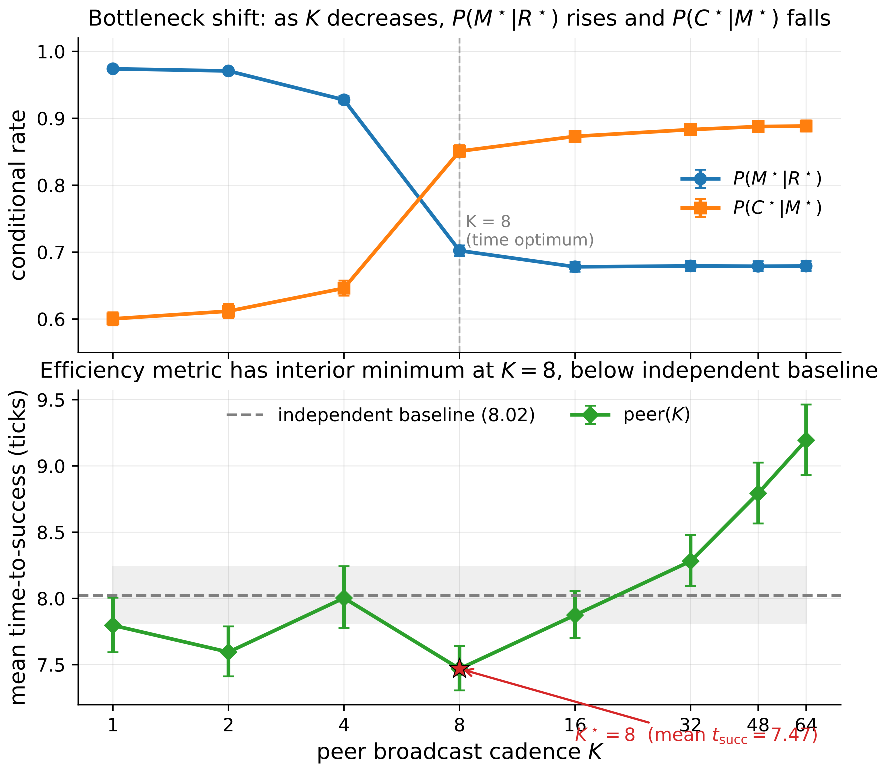

Figure 3: Bottleneck shift across peer cadences. $P(M^\star|R^\star)$ falls and $P(C^\star|M^\star)$ rises as K grows. The practical efficiency metric, mean time-to-success, is minimized at $K{=}8$.

Staleness. How old evidence decays before it influences retrieval.

The minimal CSM uses peer broadcast at $K{=}8$ and overlays three rules:

Merge. Per-place score equals own evidence at weight 1 plus peer evidence at weight $\mathrm{trust}_i \cdot e^{-\alpha\cdot\mathrm{age}}$, with $\alpha = 0.05$. Inclusion threshold $\tau = 0.30$.

Trust. Per-peer EMA on retrieval-success consistency. Peers whose latest snapshot supports the locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

Staleness. Each broadcast snapshot is timestamped; merge weight decays as $e^{-\alpha\cdot\mathrm{age}}$. Snapshots older than $10K$ ticks are pruned.

The implementation is a drop-in replacement for fixed-cadence peer memory: `experiments/collective_semantic_memory/csm_memory.py`.

5.1 Results

We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on scarcity cells $(N, T){\in}\{(3, 2), (5, 3), (8, 5)\}$, three layouts, three hazard densities, and 20 seeds: 3,240 runs total.

CSM occupies a third regime. Fast broadcasts have excellent materialization but poor completion; slow broadcasts have the reverse. CSM keeps both conditional links above 0.78. The absolute success gain is modest, $+1.7$ to $+4.0$ percentage points, but the stage decomposition shows why it is meaningful: the architecture changes the joint M$\times$C operating point rather than merely moving along the fixed-cadence curve.

Table 1: Minimal CSM vs fixed-cadence peer broadcast on the scarcity benchmark. CSM's lower 95% bootstrap CI bound exceeds every peer-K upper CI.

Architecture	$P(M^\star \mid R^\star)$	$P(C^\star \mid M^\star)$	Success rate (95% CI)
peer (K=1)	0.991	0.586	0.577 [0.572, 0.586]
peer (K=4)	0.955	0.623	0.581 [0.572, 0.590]
peer (K=8)	0.730	0.871	0.599 [0.591, 0.607]
peer (K=16)	0.697	0.892	0.597 [0.589, 0.606]
peer (K=64)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	**0.782**	**0.825**	**0.617 [0.610, 0.624]**

6 Categorical BDI reading of collective semantic memory

This section gives a compact theoretical reading of the water-search example in terms of belief, desire, and intention. The goal is not to introduce a new planner, but to make explicit the mathematical structure already used by the collective semantic memory architecture.

We assume a finite set of agents $A = \{1, \dots, N\}$, a shared spatial basis X, and a shared semantic alphabet Σ. In the present grid-world implementation, X is the common coordinate frame and Σ contains tags such as `water_source`, `water_candidate`, `hazard_edge`, and `open`. This is an explicit modelling assumption. The agents may have different observations, different memories, and different confidence values, but they interpret places in a common coordinate framework and express semantic evidence in a common tag alphabet. Without these two assumptions, peer merge would require a separate place-alignment and symbol-alignment problem, which is outside the scope of the current experiment.

At time t, agent i maintains a local semantic context $\mathcal{K}_t^i = (G_t^i, \Sigma, I_t^i)$, where G_t^i is the set of place records stored in the agent's private memory, and

$$I_t^i \subseteq G_t^i \times \Sigma$$

is the thresholded incidence relation saying that place $g \in G_t^i$ carries tag $\sigma \in \Sigma$ with sufficient confidence. The graded implementation uses tag confidences and weighted scores; the crisp relation I_t^i is the thresholded skeleton needed for the theoretical description.

This formal context induces a Galois connection between sets of tags and sets of places:

$$\mathrm{ext}_t^i(B) = \{g \in G_t^i : \forall \sigma \in B,\ (g, \sigma) \in I_t^i\}, \qquad \mathrm{int}_t^i(S) = \{\sigma \in \Sigma : \forall g \in S,\ (g, \sigma) \in I_t^i\}.$$

For a water-search task, the desired semantic predicate is $B_{\mathrm{water}} = \{\texttt{water\_source}\}$, or, in the implemented weighted form, a predicate combining positive evidence for `water_source`, `water_candidate`, and `water_nearby` with penalties for `hazard_edge` or nearby risk. Retrieval succeeds for agent i at time t exactly when

$$\mathrm{ext}_t^i(B_{\mathrm{water}}) \neq \varnothing.$$

Thus the retrieval stage is not a coordinate lookup. It is the test that the queried semantic meaning already has a non-empty extent in the agent's realized memory.

We can now define the BDI components.

Belief. The belief state of agent i is the structured semantic memory induced by $\mathcal{K}_t^i$. Equivalently, it is the concept lattice $L_t^i = \mathrm{Concepts}(\mathcal{K}_t^i)$, whose nodes pair extents of places with intensions of tags. A water belief is not the proposition "there is water at coordinate x." It is the region of L_t^i whose intension contains the water predicate. If this region is empty, the agent does not merely choose badly; the relevant semantic object is absent from its current memory.

Desire. Desire is a state-conditioned semantic predicate over the belief lattice. In the water example, the agent's internal state contains a thirst variable. The active desire can be written as

$$D_t^i = \begin{cases} B_{\mathrm{water}}, & \text{if } \mathrm{thirst}_t^i \geq \theta_{\mathrm{on}}, \\ B_{\mathrm{goal}}, & \text{otherwise.} \end{cases}$$

Thus desire is not an additional metaphysical object. It is the rule that selects which semantic predicate should be queried from the current belief structure.

Intention. Intention is the commitment to one materialized candidate. Once desire has selected a predicate and retrieval has produced a non-empty extent, the planner applies a satisficing choice function

$$\chi_t^i : \mathcal{P}(G_t^i) \to G_t^i \cup \{\perp\}, \qquad g_t^{i,\star} = \chi_t^i\big(\mathrm{ext}_t^i(D_t^i)\big),$$

where $\perp$ denotes failure to materialize a target. In Q/R/M/C terms, desire determines the query $Q^\star$, belief supports retrieval $R^\star$, and intention corresponds to materialization $M^\star$. Completion $C^\star$ is downstream of this semantic structure: it depends on whether the physical controller reaches the locked target and whether the target remains available under multi-agent occupancy.

This gives a compact categorical reading. Let $\mathbf{Ctx}_\Sigma$ be the category whose objects are semantic contexts (G, Σ, I) over the fixed alphabet Σ, and whose morphisms preserve semantic incidence. A local memory trajectory for agent i is a time-indexed diagram

$$\mathcal{K}_0^i \to \mathcal{K}_1^i \to \cdots \to \mathcal{K}_t^i \to \mathcal{K}_{t+1}^i \to \cdots$$

in $\mathbf{Ctx}_\Sigma$. Each arrow is induced by observation and memory update. When new evidence is consolidated, the formal context changes; therefore its concept lattice changes. This is the order-theoretic content of memory growth.

The H-MDP view is obtained by treating the high-level state of agent i as

$$h_t^i = (L_t^i,\, D_t^i,\, g_t^{i,\star},\, s_t^i),$$

where L_t^i is the belief lattice, D_t^i is the active desire predicate, $g_t^{i,\star}$ is the current intention, and s_t^i is the low-level physical state, including position, risk budget, thirst, and energy. The transition $h_t^i \to h_{t+1}^i$ combines two coupled processes: a semantic-memory update $\mathcal{K}_t^i \to \mathcal{K}_{t+1}^i$ and a physical control transition $s_t^i \to s_{t+1}^i$. The useful point is that the semantic part and the physical part fail in different ways. An empty extent $\mathrm{ext}_t^i(D_t^i) = \varnothing$ is a belief/retrieval failure. A non-empty extent with no stable lock is a materialization failure. A successful lock that cannot be reached is a completion failure.

We now lift the construction to the collective case. At broadcast times, agent i receives snapshots from peers. For the receiver i, these snapshots form a diagram of local memory objects

$$\mathcal{D}_t^i = \big(\mathcal{K}_t^i,\, \mathcal{K}_t^{j_1}, \ldots, \mathcal{K}_t^{j_m}\big),$$

together with alignment maps induced by the shared coordinate basis and semantic tag compatibility. The merged view used by agent i is a colimit-like consensus object

$$\widetilde{\mathcal{K}}_t^i = \mathrm{Merge}_{\mathrm{trust,stale}}\big(\mathcal{D}_t^i\big).$$

This notation is intentionally modest. The implementation does not claim to solve general semantic alignment. It computes a practical merge over aligned place records, with peer evidence weighted by trust and staleness. A typical weight has the form

$$w_{j\to i}(t) = \mathrm{trust}_{j\to i}(t)\, \exp\big(-\alpha\, \mathrm{age}_j(t)\big),$$

so the merged incidence strength for a tag σ at a place g can be written schematically as

$$\widetilde{I}_t^i(g,\sigma) = I_t^i(g,\sigma) + \sum_{j\neq i} w_{j\to i}(t)\, I_t^j(g,\sigma),$$

followed by thresholding or weighted scoring during retrieval. This separates three control surfaces: *merge* (what evidence refers to the same place), *trust* (whose evidence should count more), and *staleness* (how quickly old evidence loses force).

The same BDI definitions then apply to the merged context $\widetilde{\mathcal{K}}_t^i$. The agent's collective belief is the lattice $\widetilde{L}_t^i = \mathrm{Concepts}(\widetilde{\mathcal{K}}_t^i)$, its desire is still a semantic predicate D_t^i, and its intention is now selected from $\mathrm{ext}_{\widetilde{\mathcal{K}}_t^i}(D_t^i)$ rather than from its private extent alone.

This explains why communication cadence produces a bottleneck shift. Faster broadcast increases the freshness of peer evidence. This tends to enlarge or improve the water extent $\mathrm{ext}_{\widetilde{\mathcal{K}}_t^i}(B_{\mathrm{water}})$, so agents can materialize targets more reliably; in Q/R/M/C terms, this raises $P(M^\star \mid R^\star)$. But the same fresh shared belief also synchronizes intentions. Several agents may lock onto the same scarce water source. The memory is semantically accurate, but the physical target is now contested. This lowers $P(C^\star \mid M^\star)$. The bottleneck is therefore physically grounded. It is not a formal artifact of the decomposition: it arises because semantic belief is shared through memory, while completion is constrained by occupancy and scarce resources. Collective semantic memory must therefore regulate not only how much information is shared, but also how it is merged, whose evidence is trusted, and when stale evidence should stop influencing intention formation.

In this sense, the "collective" is not a central mind and not a global map. It is the stabilized family of peer-indexed belief updates and intention-forming rules that remains invariant under communication. A population has acquired a collective water-search convention when the same ostensive evidence, such as a peer marking a water-like place, induces compatible updates across agents' local merged contexts and leads to coordinated, non-colliding intentions. The category-theoretic role of the model is to make this invariant explicit: individual agents are local indexed instances of the same semantic-memory process; communication exposes partial peer contexts through snapshots; collective memory is the merge object that makes those contexts jointly usable without collapsing them into a single undifferentiated global memory.

6.1 Colimit merge and its invariants

The merge object above can be described more precisely. Working in a timestamp- and provenance-enriched version of $\mathbf{Ctx}_\Sigma$, a snapshot merge from agents $1, \ldots, N$ is a diagram of local memory objects plus alignment maps induced by the place-identity relation. The consensus object $\widetilde{\mathcal{K}}_t^i$ is the colimit of this diagram, implemented concretely by the union-find merge over aligned concepts. This reading gives three useful constraints.

1. **Order independence.** The merged object is unique up to isomorphism, so the result is not an artefact of processing agent snapshots in a particular order.

2. **Chaining is expected.** If concept A aligns with B and B aligns with C, union-find performs the transitive closure. Over-merging is therefore not a bug in the data structure; it is the cost of the chosen equivalence relation.

3. **Decay is not a morphism.** Staleness is kept as a weight field outside the alignment morphisms. Otherwise age-then-merge and merge-then-age need not commute, breaking the clean interpretation of broadcast snapshots as one diagram.

The model does *not* answer which perceptual features should identify two places. In the current grid substrate, identity still depends partly on coordinate keys and semantic tag similarity. Removing that coordinate anchor requires a separate place-identity experiment: agents must infer shared objects from local signatures rather than from global (x, y) keys.

7 Limitations and next experiments

Grid substrate. The current result is intentionally controlled: all experiments are in grid worlds. A continuous-state substrate such as VMAS, Habitat, ProcTHOR, or MiniWorld is the next portability step.

Coordinate anchor. The present implementation can use shared coordinates for atomic place keys. A stronger version of the paper should remove that anchor and add local signatures for place identity. The categorical merge story remains valid only after the alignment relation has been defined; it does not define the relation for us.

Minimal CSM. The CSM rules are deliberately simple. Adaptive cadence K_t, intent-conditioned trust, and selective consolidation of under-represented tags are the next architecture variants.

Learned baselines. CommNet-style learned communication is a useful comparison, but the more relevant future baseline is an explicit memory LLM or vector-memory multi-agent system measured on the same task.

8 Conclusion

Collective memory is not monotone. In scarce multi-agent navigation, instant information sharing can reduce completion by causing agents to converge on the same targets. Peer cadence exposes the trade-off, but cadence alone is a one-dimensional control surface. A collective semantic memory should make merge, trust, and staleness explicit. The minimal CSM does exactly that and moves the system to a new operating point: neither fast-broadcast M-saturation nor slow-broadcast C-saturation, but a more balanced M×C profile with higher success than every fixed peer cadence in the scarcity benchmark.

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A Reproducibility commands

- Main cadence sweep: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended cadences: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- Q/R/M/C analysis: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Minimal CSM benchmark: `python-mexperiments.collective_semantic_memory.run_csm_benchmark--out_dirtmp/csm_benchmark`
- MiniGrid portability sweep: `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`

B Asset and license note

The experiments use MiniGrid and MiniWorld (Chevalier-Boisvert et al., 2023). Locally implemented benchmark and memory code will be released under the MIT License as part of the artifact package.

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